

VASP Ensemble and Force-Provenance Certificate Specification

Stage: 11E-ENS1

Package target: mdstats 0.20.18a0

Status: implemented

1. Purpose

Stage 11E-ENS1 converts the exact source evidence reconstructed by ENS0 into a signed, source-bound interpretation of the realized molecular-dynamics control mode. It determines the control-inferred dynamics mode, ensemble, propagator, thermostat, barostat, cell control, bias/constraint evidence, force-provider provenance, initial-velocity provenance, and continuation provenance.

ENS1 does not evaluate whether the trajectory actually conserved energy, reached stationarity, or is thermodynamically admissible for a PMF. Those questions belong to STAT0-STAT2.

The VASP SYSTEM field, file names, directory names, and human labels are non-authoritative comments. They must never influence the certificate.

2. Inputs and source identity

The primary API is:

```
certify_vasp_simulation_controls(  
    source: str | Path | VaspSourceControlBundle,  
    *,  
    companion_files: Mapping[str, str | Path] | None = None,  
) -> SimulationControlCertificate
```

The output is bound to all three ENS0 identities:

- SourceTrajectoryBundleIdentity.signature;
- VaspSourceControlBundle.signature;
- VaspRunControls.signature.

A certificate is invalid if any source signature changes.

3. Persistent records

3.1 SimulationControlComponent

Each component has an independent status:

- resolved;
- unresolved;
- conflicting;
- not_applicable.

It records a kind, active state, typed parameters, source evidence, and notes. Missing companion evidence is never converted into affirmative absence.

3.2 SimulationControlDecision

Every applied inference rule is recorded with a stable rule identifier, outcome, decisive control evidence, and explanatory notes.

3.3 SimulationControlCertificate

The certificate contains:

- control-inferred dynamics mode;

- ensemble and inference status;
- propagator;
- thermostat and friction parameters;
- barostat;
- cell-control mode;
- bias and constraint evidence;
- force-provider and applied-force provenance;
- initial-velocity evidence;
- continuation/restart evidence;
- unresolved reasons and warnings;
- complete decision trace;
- canonical SHA-256 signature.

4. VASP inference policy

The initial policy version is `mdstats.ensemble-inference-policy.v1+vasp-wiki-2026-06`.

4.1 Dynamics selection

- `IBRION = 0` selects molecular dynamics.
- Other `IBRION` values are classified as non-MD for this certificate; an MD ensemble is then `not_applicable`.

4.2 Andersen dynamics

For `MDALGO = 1`:

- `ANDERSEN_PROB = 0` -> NVE, thermostat disabled;
- `ANDERSEN_PROB > 0` -> NVT with Andersen collisions;
- missing or invalid collision probability -> unresolved.

4.3 Nose-Hoover family

For `MDALGO = 0 | 2`:

- `SMASS = -3` -> NVE, thermostat disabled;
- `SMASS = -2` -> constant-velocity path, not an equilibrium ensemble;
- `SMASS = -1` -> deterministic velocity-rescaled temperature schedule;
- `SMASS >= 0` -> NVT with Nose-Hoover thermostat;
- missing or unsupported `SMASS` -> unresolved.

When the thermostat is disabled, friction is recorded as `not_applicable`, not as zero and not as unknown.

4.4 Langevin and variable-cell dynamics

For `MDALGO = 3`:

- fixed cell (`ISIF <= 2`) and positive atomic `LANGEVIN_GAMMA` -> NVT;
- fixed cell and all-zero atomic friction -> NVE;
- variable cell (`ISIF = 3`), positive atomic and lattice friction -> NpT;
- variable cell, zero atomic and lattice friction -> NpH;
- mixed or incomplete friction controls -> conflicting/unresolved.

`LANGEVIN_GAMMA` and `LANGEVIN_GAMMA_L` retain their source units of ps^{-1} . `PMASS` and `PSTRESS` are retained as control parameters rather than silently interpreted by `ENS1`.

4.5 Other VASP propagators

- `MDALGO = 4` -> Nose-Hoover-chain NVT;

- MDALGO = 5 -> CSVR NVT;
- MDALGO = 13 -> multiple independently thermostatted subsystems and a nonstandard multi-thermostat classification;
- unsupported algorithms remain unresolved and do not fall back to a user label.

5. Cell and barostat provenance

- ISIF <= 2 -> fixed cell;
- ISIF = 3 -> variable volume and shape;
- other variable/partial cell modes remain unresolved until a dedicated policy exists.

A fixed cell has no active barostat. For MDALGO = 3, ISIF = 3 selects the Parrinello-Rahman variable-cell family; the lattice friction determines whether it is thermostatted.

6. Bias and constraint provenance

A bound and parsed ICONST file is authoritative:

- status 0 -> constrained coordinate;
- status 3, 4, 5, or 8 -> forcefield, step bias, metadynamics, or harmonic bias evidence;
- status 7 -> monitoring only.

Metadynamics controls such as HILLS_H, HILLS_W, and HILLS_BIN provide positive bias evidence but do not replace the collective-variable definition.

If ICONST, REPORT, PENALTYPOT, or related companion evidence is not provided, ENS1 reports unresolved. It does not assert that bias and constraints were absent.

7. Force provenance

ENS1 classifies the force provider from effective source controls:

- standard VASP DFT/Hellmann-Feynman force path;
- VASP MLFF prediction mode;
- VASP on-the-fly hybrid MLFF/DFT training mode;
- unresolved MLFF mode.

Force-array completeness is recorded independently from applied-force provenance. Complete arrays do not prove that bias or constraint contributions were absent. Therefore force provenance may remain unresolved even when the ensemble and source force provider are resolved.

8. Initial velocities and continuation

Native velocity records are affirmative evidence. When native velocities are absent but the first ionic kinetic energy is positive, ENS1 records nonzero initial kinetic energy while leaving its source unresolved.

A continuation claim requires a bound parent/restart source. A nonzero initial kinetic energy with default or inconsistent TEBEG values may be reported as consistent with continuation or external initialization, but it is not proof of lineage.

9. Execution policy

An unresolved or conflicting ensemble blocks only methods that require a resolved ensemble interpretation. Descriptive coordinate, density, and structural analyses may continue if later STAT0 hard-integrity checks pass.

Bias, constraint, or force-provenance uncertainty does not invalidate the raw trajectory. It blocks only the scientific interpretation that needs the missing provenance, such as an unqualified physical mean-force claim.

10. Real Na-LTA acceptance fixture

The supplied 1,500-step `vasprun.xml` must resolve as:

- VASP molecular dynamics;
- `MDALGO = 2`, `SMASS = -3`, `ISIF = 2`;
- fixed-cell NVE;
- Nose-Hoover-family propagator with thermostat disabled;
- thermostat friction `not_applicable`;
- no barostat;
- complete DFT force arrays;
- bias and constraint status unresolved without companion evidence;
- nonzero initial kinetic energy with unresolved velocity source;
- continuation/external initialization possible but not proven;
- misleading `SYSTEM = ... NVT ...` ignored.

11. Tests

Focused tests must cover:

- misleading `SYSTEM` label rejection;
- Nose/Andersen NVE and NVT branches;
- Langevin NVE, NVT, NpT, and NpH branches;
- `SMASS = -1` and `-2` driven modes;
- bound `ICONST` constraint and bias classification;
- force-provider and force-completeness separation;
- immutable serialization and tamper detection;
- VASP frame-reader metadata integration;
- exact real-source replay.

12. Primary VASP references

- VASP Wiki, `MDALGO`: <https://vasp.at/wiki/MDALGO>
- VASP Wiki, `SMASS`: <https://vasp.at/wiki/SMASS>
- VASP Wiki, NVE ensemble: https://vasp.at/wiki/NVE_ensemble
- VASP Wiki, NVT ensemble: https://vasp.at/wiki/NVT_ensemble
- VASP Wiki, NpT ensemble: https://vasp.at/wiki/NpT_ensemble
- VASP Wiki, NpH ensemble: https://vasp.at/wiki/NpH_ensemble
- VASP Wiki, `LANGEVIN_GAMMA`: https://vasp.at/wiki/LANGEVIN_GAMMA
- VASP Wiki, `LANGEVIN_GAMMA_L`: https://vasp.at/wiki/LANGEVIN_GAMMA_L
- VASP Wiki, `ICONST`: <https://vasp.at/wiki/ICONST>
- VASP Wiki, metadynamics calculations: https://vasp.at/wiki/Metadynamics_calculations